

CLOSED LOOP CONTROLLER FOR BI-DIRECTIONAL WEINBERG DC-DC CONVERTER FOR SPACE SATELLITE APPLICATION

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Abstract—In this paper, a non-isolated bi-directional Weinberg circuit topology for battery charge-discharge in space satellite applications is presented. In order to provide a better dynamic response to the DC bus voltage regulation during the discharge mode of operation, a double loop control is employed in the analog domain. Adding an additional switch to operate in synchronous mode mitigate thermal stress on the switches, thereby enhancing the converter's performance and efficiency. To verify the performance of the converter and its control scheme in the discharge mode of operation, simulation was carried out in PSIM and the results are provided. Finally, the experimental results from a 500 W prototype are provided in order to validate the design.

Index Terms—Bi-directional Converter, Battery, Weinberg Converter, Double loop control

I. INTRODUCTION

Advances in space technology demand an efficient power supply system. However, these systems take up more space within the spacecraft and increase launch expenses. Additionally, increasing spacecraft power density offers considerable practical advantages [1] [2]. Employing separate regulators for charging and discharging the batteries within satellite power systems may introduce inefficiencies, such as reduced power density and circulating losses arising from energy interchange between the regulators. Conversely, integrating these functionalities into a singular converter architecture can significantly streamline the power system. This integration not only reduces the overall size and complexity of the satellite's power management infrastructure but also offers a cost-effective solution by eliminating the need for multiple discrete components, thus enhancing the efficiency and reliability of the satellite's energy storage and distribution mechanisms.

The choice of converter for battery charging and discharging largely depends on the battery voltage. Typically, a buck-derived topology is employed for the battery charging regula-

tor, while a boost-derived topology is for the battery discharge regulator. Among the options for battery discharge regulators, the Super-boost, Add-on Smart, and non-isolated Weinberg converter typologies [3] are considered. However, the non-isolated Weinberg topology stands out as the optimal choice, because the Super-boost converter's plants transfer function is of 4th order and it introduces complexity into its design. Additionally, its peak current and frequency are double the Weinberg converter's, resulting in increased conduction losses and switching losses. The Add-on Smart is similar to the Weinberg converter but includes an additional buck stage for PWM regulation, leading to efficiency losses [4].

The Weinberg topology is derived from the Buck and current-fed Push-pull topology [5]. There are two types of Weinberg converters namely, isolated and non-isolated. The non-isolated Weinberg converter topology is increasingly utilized in spacecraft power conditioning unit [6]. A bidirectional non-isolated Weinberg converter in [8] has increased power density and efficiency. However, during the discharging mode of operation, the diode at the DC bus side conducts for a long time, resulting in increased losses and thermal stress.

In this paper, a battery discharge regulator(BDR) for a Bidirectional Synchronous Weinberg converter(BSWC) is presented and is shown in Fig 1. In this converter configuration, the diode at the DC bus side is replaced with power MOSFET Q_3 . During the discharge mode, switch Q_3 operates in synchronous mode with switches Q_1 and Q_2 as shown in Figure 2 and switch Q_3 alone during charging mode of operation. Replacing the diode with a MOSFET facilitates bidirectional current flow, making the converter suitable for both battery charging and discharging applications. This converter interfaces a DC bus with a voltage of 100 V to a battery that operates within a voltage range of 66 V to 96.6 V.

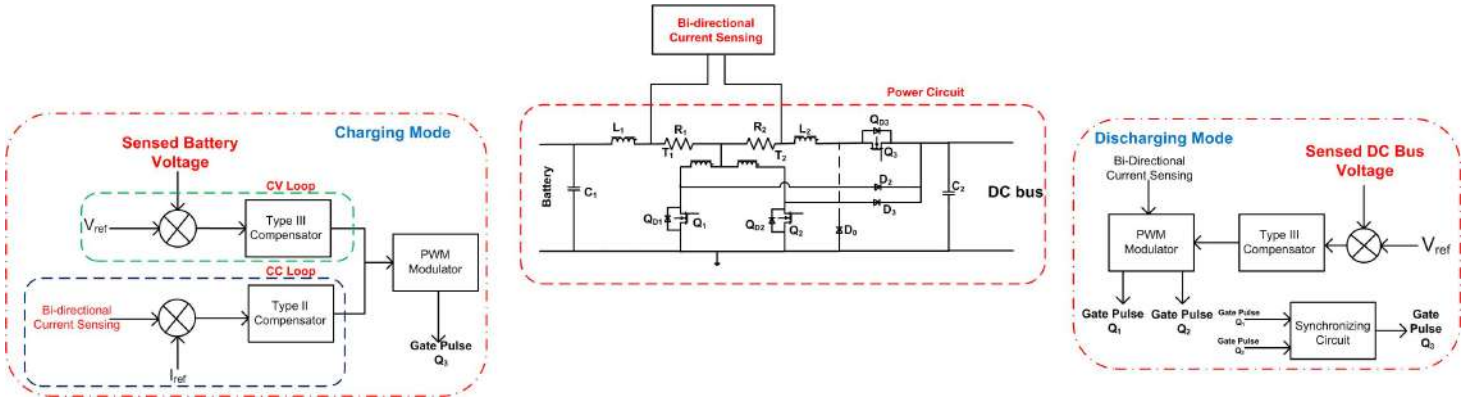


Fig. 1: Bi-Directional Weinberg Converter with Charge and Discharge Regulation

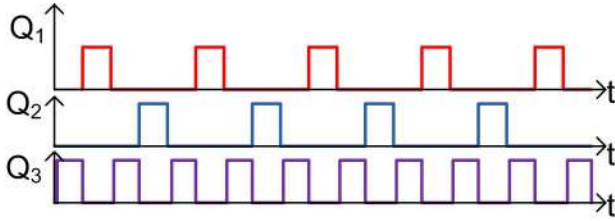


Fig. 2: Gate Pulses of Switches Q_1 , Q_2 and Q_3

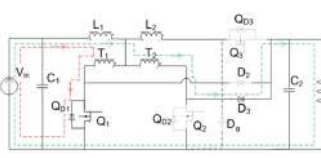


Fig. 3: BSWC discharge Mode 1

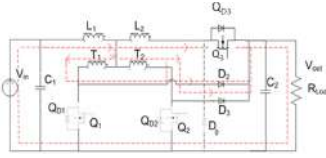


Fig. 4: BSWC discharge Mode 2

II. THE CHARGE-DISCHARGE MODES OF OPERATION

A. Discharge mode

When the voltage at the DC bus falls below the nominal voltage of 100 V, the discharge mode begins. In this mode, the gate pulses for Q_1 and Q_2 have a phase shift of 180 degrees, while Q_3 operates in synchronous with both Q_1 and Q_2 as shown in Figure 2.

1) Mode 1: [Q_1 -ON, Q_2 -OFF, D_2 -OFF, D_3 -ON]

When the discharge mode begins, current passes through both inductor L_1 and the primary winding of the transformer T_1 as switch Q_1 is turned ON. Since the turn's ratio of the transformer is kept at 1, the current passing through the secondary matches that of the primary ($i_{Q1} = i_{T1} = i_{T2}$). While switch Q_2 remains inactive in this mode, the current in the transformer's secondary will be directed through diode D_3 and to the output. During this condition, the voltage at the midpoint of the transformer (V_{CT}) equals half of the output voltage ($V_{CT} = \frac{V_{OUT}}{2}$), causing the body diode of switch Q_3 to become reverse biased.

Neglecting magnetizing and leakage inductances, the input current is expressed as $i_{in} = i_{Q1} + i_{D3}$ and alternatively as $i_{in} = i_{Q1} = i_{D3}$, resulting in $i_{in} = 2i_O$.

2) Mode 2: [Q_1 -OFF, Q_2 -OFF, D_2 -ON, D_3 -ON]

In this mode, switch Q_1 is turned off. The magnetizing inductance of transformer L_{Tm} maintains the same current equal to the primary current i_{T1} of the transformer through D_2 , thereby resetting the transformer magnetically. Consequently, the body diode of switch Q_3 becomes forward-biased, allowing the inductor current to gradually increase until it reaches the rated output current.

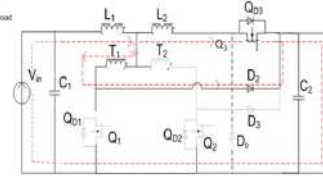


Fig. 5: BSWC discharge Mode 3

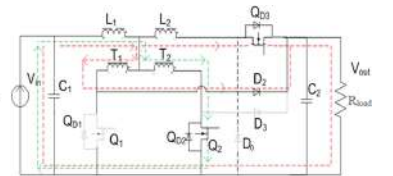


Fig. 6: BSWC discharge Mode 4

3) Mode 3: [Q_1 -OFF, Q_2 -OFF, Q_3 -ON, D_2 -OFF]

In Mode 3, switch Q_3 is engaged, synchronizing its operation with that of Q_1 and Q_2 . This switching ensures that during the dead time, the inductor current initially flows through the diode and then through switch Q_3 once it turns ON, owing to the lower voltage drop across the switch than the diode. The lower voltage drop across the switch effectively minimizes power dissipation, enhancing overall efficiency and optimizing thermal management.

The calculation of power dissipation in the synchronous switch involves accounting for conduction losses attributed to $R_{DS(on)}$ and the body diode, as detailed in equation 1.

$$P_{Q3} = \left(\frac{I_o^2 \times R_{DS(ON)}}{1 + D} \right) \times \left(V_{FD} \times I_o \times 2 \times t_{delay} \times f_{sw} \right) \quad (1)$$

In this equation, D represents the duty ratio, and V_{FD} denotes the forward voltage drop of the body diode associated with switch Q_3 .

4) *Mode 4*: [Q_1 -OFF, Q_2 -ON, Q_3 -OFF, D_3 -OFF]

During mode 4, switch Q_2 is turned ON, while Q_1 and Q_3 are turned OFF. Due to the inductance of the transformer, sudden changes in current through its windings are prevented. As a result, the current through Q_2 and D_2 gradually rises, causing a gradual decrease in the current through the body diode of Q_3 .

B. Charge mode

The charging mode begins when the DC bus voltage reaches 101 V and the battery voltage drops below 96 V. In this mode, power is transferred from the DC bus to the battery using a conventional buck converter topology. This circuit consists of an inductor L and a freewheeling diode D_0 , which together perform the necessary voltage step-down to facilitate charging. Notably, transistors Q_1 and Q_2 are not operational during this phase, as their switching is not required for the power conversion.

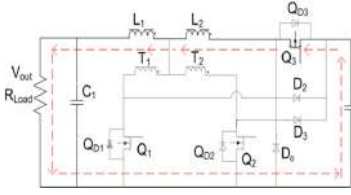


Fig. 7: BSWC charge Mode 1

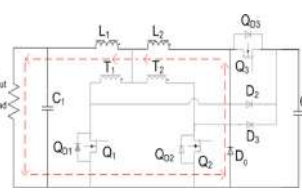


Fig. 8: BSWC charge Mode 2

1) *Mode 1*: [Q_3 -ON, D_0 is OFF]

In mode 1, switch Q_3 is turned ON. The coupled inductor begins charging, hence charging the battery. During this phase, diode D_0 operates in a reverse bias state.

2) *Mode 2*: [Q_3 -OFF, D_0 -ON]

In mode 2 of operation, switch Q_3 is deactivated. Due to the inductor's reluctance to sudden changes in current, it discharges through diode D_0 .

III. SMALL SIGNAL MODEL

The transfer function of the BSWC is obtained through small-signal analysis. The resulting transfer functions for the discharge and charge modes are shown in Equations 2 and 3, respectively.

$$\frac{V_{out}(s)}{d(s)} = \frac{V_{in}}{(4LC_d(S^2 + \frac{1}{R_d C_d}S + \frac{1}{4LC_d}))} \quad (2)$$

$$\frac{V_{out}(s)}{d(s)} = \frac{V_{in}}{(LC_c(S^2 + \frac{1}{R_c C_c}S + \frac{1}{LC_c}))} \quad (3)$$

Where R_d and R_c are equivalent resistance during discharging and charging process respectively. C_d is equivalent output capacitor during discharge process and C_c equivalent output capacitor of battery during charging process. According to equations 2 and 3 both the discharging and charging modes share an identical small signal model, which is of second-order without any right half plane zeros or poles. This similarity simplifies control and enhances stability compared to

conventional non-isolated bidirectional converters [8], which typically exhibit a right half plane zero in their boost mode operational topology.

A type 3 compensator is designed to achieve the required phase boost of 141 degrees for the discharging mode and 137 degrees for the charging mode. The closed-loop reference to output response bode plots obtained after compensation is shown in Figure 9 and Figure 10. For the discharge mode, a phase margin of 60 degrees and a gain margin of 14 are achieved. In the charging mode, a phase margin of 60 degrees and a gain margin of 24.9 are obtained.

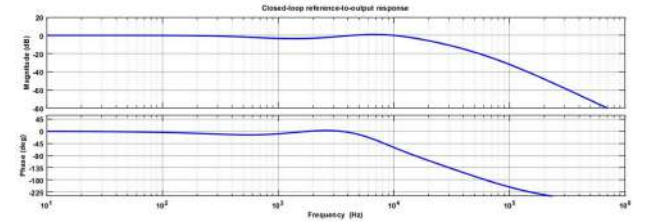


Fig. 9: Bode plot of BSWC during discharge mode

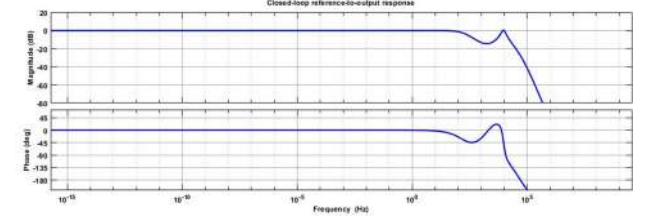


Fig. 10: Bode plot of BSWC during charge mode

This paper focuses on the discharge mode of operation and DC bus voltage control by regulating the DC bus to 100 V, while considering a battery that operates within a range of 66 to 96.6 V.

IV. DOUBLE LOOP BUS VOLTAGE CONTROL

BDR is designed with a double-loop control system to regulate bus voltage as shown in Figure 11, enabling seamless power extension and connecting multiple modules in parallel. The inner current loop and an outer voltage loop form the double loop control system. Despite variations in battery voltage and external disruptions, the control system's objective is to control the bus voltage V_{out} by keeping alignment with a steady reference voltage.

Recognizing the faster dynamics of inductor current than capacitor voltages, the inner loop is designated for current control, and the outer loop is reserved for voltage control in the double-loop system. This arrangement enhances phase response and mitigates electrical stress on system components by regulating current levels, thereby stabilizing the DC bus output at 100V.

The UC3825N PWM IC is used to generate the gating pulses for switches Q_1 and Q_2 , operating based on a standard duty cycle control method. This IC controls the switching by adjusting the duty cycle as needed. To ensure the sensed DC bus voltage aligns with the voltage loop reference signal, a

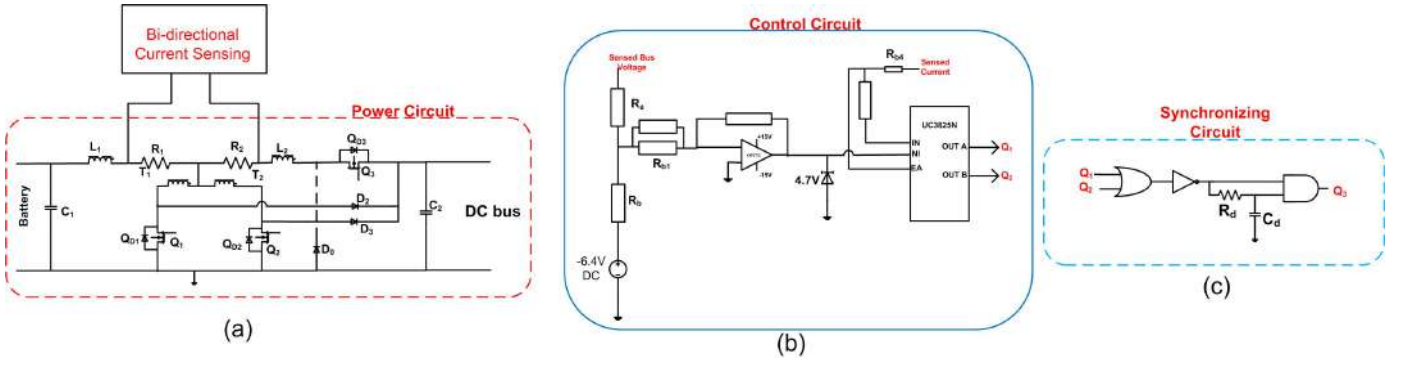


Fig. 11: Double loop DC bus voltage control of BSWC

resistive voltage divider is paired with a negative 6.4V DC bias. This setup scales the DC bus voltage to a level that the control system can handle. The configuration, as shown in Figure 11.b, generates an instantaneous error signal, allowing the system to correct any difference between the sensed and reference voltages, thus maintaining stable voltage regulation.

The reference signal for the inner current loop is generated by the outer voltage loop's Type III compensator. This compensator adjusts the voltage loop and optimizes the system's stability. Once the current loop's reference is established, an instantaneous current error signal is created by comparing this reference with the actual current measured in real-time. This error signal indicates the difference between the desired and actual current values, enabling precise control of the current within the converter.

With the combined action of the inner current and outer voltage control loops, gating pulses are produced for switches Q_1 and Q_2 , with a 180° phase shift between them. This phase difference allows the switches to operate alternately, improving power flow and efficiency. The gating signal for switch Q_3 , which operates in synchronously with Q_1 and Q_2 , is generated using the logic as shown in Figure 11.c.

A. Simulation and Experimental Results

A 500 W BSWC and its control scheme have been designed and simulated using PSIM software. Additionally, a 500 W hardware laboratory prototype has been developed, as illustrated in Figure 12. The parameters used for both simulation and experimental validations are detailed in Table I.

TABLE I: Converter Specifications and Parameters

DC Bus voltage	100V
Battery voltage	66 to 96.6V
Charging power(Buck operation)	500W
Discharging power(Boost operation)	500W
Switching frequency	100KHz
Inductor	40 μ H
Capacitor	1000 μ F

In Figure 13, the system's voltage control capabilities maintaining a constant output voltage of 100 V across a range of supply input voltages from 66 V to 96.6 V is shown. A step change in V_{in} from 66 V to 75 V causes a transient

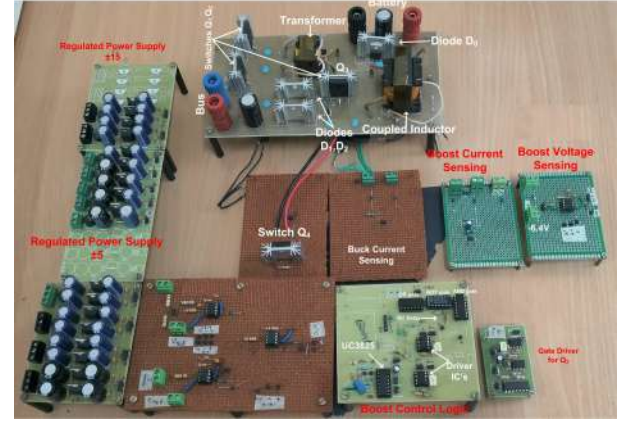


Fig. 12: Hardware Prototype of BSWC and Control

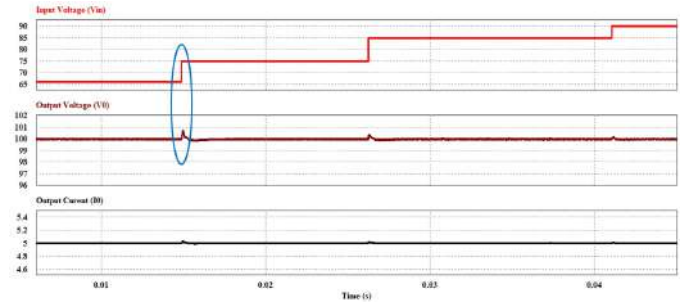


Fig. 13: Constant bus voltage at varying input voltage

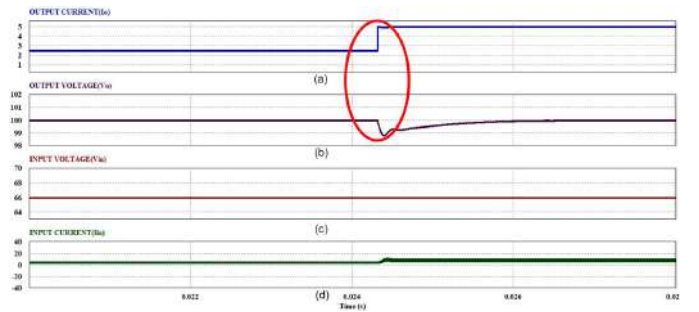


Fig. 14: Constant bus voltage with change in load

voltage variation at the output from 100 V to 100.8 V, with a settling time of 0.6 milliseconds. Similarly, when the input voltage shifts from 75 V to 85 V, the output voltage transiently varies from 100 V to 100.6 V, stabilizing in 0.4 milliseconds. Additionally, the behavior of the output current I_{out} during these transient periods is depicted in the same figure.

Figure.14 illustrates the response of output voltage regulation to changes in load. As the load varies from half load to full load, I_{out} also varies accordingly, maintaining output voltage constant at 100 V for a transient duration of 1.77 milliseconds. The variation of input current corresponding to load change is shown in Figure 14.

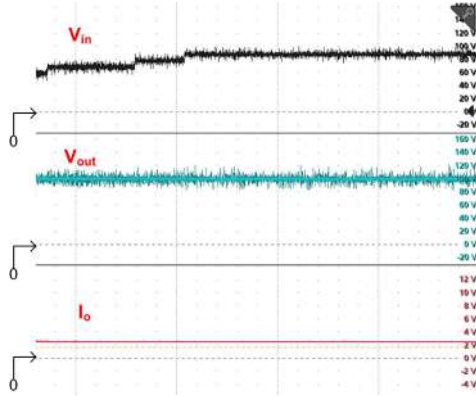


Fig. 15: Hardware prototype result of controlled DC bus voltage with 20V/Div and 10 μ s/Div

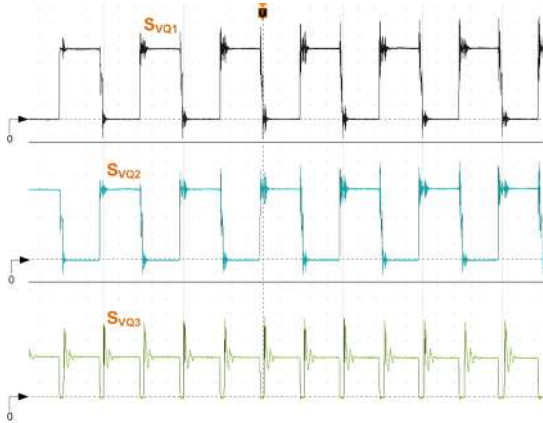


Fig. 16: Hardware result of voltage across switches with 20V/Div and 10 μ s/Div

A 500 W experimental prototype is tested to verify the design. Parameters for the experimental prototypes are given in Table I. In Figure 15, the V_{out} and I_{out} variation with respect to change in V_{in} is depicted. The input battery voltage is varied from 66 V to 96.6 V. As the battery voltage increases gradually from 66 V, the V_{out} remains regulated at 100 V. Additionally, the behavior of the I_{out} during the periods is depicted in the same figure.

The MOSFET switches Q_1 , Q_2 , and Q_3 in Figure 16 are being evaluated for their voltage characteristics. When these

switches are in operation, the voltage drop between the drain and source terminals of each switch is measured and labeled as S_{VQ1} , S_{VQ2} and S_{VQ3} respectively. It was found that the maximum voltage observed across both S_{VQ1} and S_{VQ2} is 100 volts. However, S_{VQ3} exhibited a lower maximum voltage of 60 volts during its operation

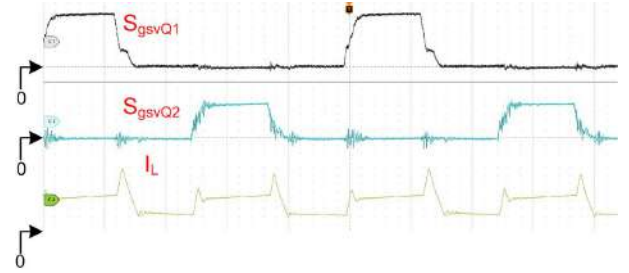


Fig. 17: Gate-source voltage of switches Q_1 and Q_2 (2.5 V/div, 3.75 V/div) and inductor current (1 A/div)

The Figure 17 depicts the gate-source voltage waveform for switches Q_1 and Q_2 , labeled as S_{gsvQ1} and S_{gsvQ2} , along with the inductor current I_L . The inductor current I_L reflects the variations in current as it responds to the alternating states of the switches.

V. CONCLUSION

The paper presents a non-isolated bidirectional synchronous Weinberg converter tailored for aerospace battery charging and discharging applications. This design replaces the traditional diode with a third switch operating in synchronous mode. This substitution significantly reduces the thermal stress on the switches, leading to improved converter performance and efficiency. The BSWC and its associated control scheme have been developed and implemented to meet the requirements of bidirectional power flow (BDR) in space satellite applications. During discharge mode, a double-loop control scheme operating in the analog domain ensures precise regulation of the DC bus voltage at 100 volts. This control strategy delivers superior dynamics in DC voltage regulation, enhancing the overall system stability. The effectiveness of the proposed converter and its control strategy has been thoroughly validated through both simulation and experimental results obtained from a 500 W prototype.

REFERENCES

- [1] Liu, Yawei and Su, Xiaobao, "High power and efficiency power combining of space TWTAs with waveguide magic-T for Satellite communication", 2015 IEEE International Vacuum Electronics Conference (IVEC), vol., no., pp.1-2, 2015.
- [2] A. Garrigos, J. A. Carrasco, J. M. Blanes and E. Sanchis-Kilders, "A power conditioning unit for high power GEO satellites based on the sequential switching shunt series regulator", MELECON 2006 - 2006 IEEE Mediterranean Electrotechnical Conference, Malaga, Spain, pp.1186-1189, 2006.
- [3] Maset, E. and Ferreres, A. and Ejea, J.B. and Sanchis-Kilders, E. and Jordan, J. and Esteve, V., "5kW Weinberg Converter for Battery Discharging in High-Power Communication Satellites", 2005 IEEE 36th Power Electronics Specialists Conference, pp.69-75, 2005.
- [4] Weinberg, A. H., "The Battery Discharge Regulator of the Galileo Satellite Test Bed-V2/A Power System using the Weinberg Topology", ESA Special Publication, vol.589, pp.11, 2005.

- [5] Li, Heng and Wu, Suliang and Jiang, Yue and Tu, Bo and Ma, Jijun and Hu, Wenbin,"Research on Battery Discharge Regulator Based on Weinberg Topology", 2020 IEEE 9th Joint International Information Technology and Artificial Intelligence Conference (ITAIC), vol.9, pp.1363-1367, 2020.
- [6] Weinberg, A.K. and Rueda Boldo, P.,"A high power, high frequency, DC to DC converter for space applications", PESC '92 Record. 23rd Annual IEEE Power Electronics Specialists Conference, vol.2, pp.1140-1147, 1992.
- [7] Qunhai, Huo and Jingyuan, Yin and Lixin, Wu and Tongzhen, Wei,"Research on a new bidirectional DC-DC topology for space applications", 2017 12th IEEE Conference on Industrial Electronics and Applications (ICIEA), pp.1686-1690, 2017.
- [8] Yang, Jae-Won and Do, Hyun-Lark,"High-Efficiency Bidirectional DC-DC Converter With Low Circulating Current and ZVS Characteristic Throughout a Full Range of Loads", IEEE Transactions on Industrial Electronics, pp.3248-3256, 2014.